

Nasik Sami Khan

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Professional Summary:

AI Software Developer and Master's in Computer Science graduate with **over 3 years of experience** in applied artificial intelligence, data engineering, and software development across healthcare, telecom, and research domains. Proven success in designing and deploying **LLM-based and RAG-driven AI systems**, optimizing model performance from research to production, and developing **data pipelines and predictive models**. Experienced in integrating **AI solutions into production-grade software**, ensuring scalability, reliability, and regulatory compliance. Strong background in **Machine Learning, Deep Learning, Generative AI, and cloud-native technologies**, with published research in AI for telecom and network intelligence. Passionate about **bridging advanced AI innovation with real-world impact** through data-driven engineering and continuous learning.

Technical Skills:

- **Programming Languages:** Python, R, C#, C++, Java, JavaScript
- **AI/ML Frameworks & Libraries:** LangChain, LangGraph, AutoGen, PyTorch, TensorFlow, Scikit-learn, MLflow
- **Software Development frameworks & Libraries:** ASP.NET (MVC, Core, WebForms), Entity Framework, Dapper
- **Tools & Technologies:** Docker, RabbitMQ, Kubernetes, Jenkins, Tableau, Power BI, Excel VBA
- **Databases:** MySQL, Oracle, Databricks, Qdrant, ChromaDB, FAISS, Pinecone
- **Developer Tools:** Git, Visual Studio Code, Postman
- **Cloud & DevOps Tools:** AWS (Lambda, S3, EC2, DynamoDB), GCP

Professional Experience:

BDM Healthware Inc.

(Sep 2025 – Present)

AI Software Developer

Saskatoon, Canada

- **Lead AI projects** within R&D to **design and deploy** healthcare solutions for external clients.
- **Build and optimize AI/ML pipelines** for data preprocessing, model training, and scalable deployment.
- **Collaborate with clinical experts, product teams, and engineers** to translate healthcare requirements into AI-powered features improving patient care workflows.
- **Research and implement state-of-the-art AI techniques**, ensuring secure, ethical, and compliant model deployment.
- **Mentor team members** about **AI architectures and processes** to ensure maintainability and knowledge continuity.

University of Regina

Graduate Research Assistant: Networking Lab at UofR. Supervised by Dr. Nashid Shahriar.

(Jan 2023 – May 2025)

- Published **four peer-reviewed** papers in journals and conferences and another two are currently under review.
- Led **LLMOps projects** using **RAG** and **BERT pipelines** for **telecom incident classification** and improved F1-score by **14%**.
- Developed a **hierarchical intrusion detection system** (CNN + Random Forest) with a **92% F1-score**.
- **Runner-up** position at "[Intrusion and Vulnerability Detection in SDN](#)" challenge by ITU AI/ML in 5G.
- **Silver Rank** at "[Specializing Large Language Models for Telecom Networks](#)" challenge by ITU AI/ML in 5G.
- **Mentored junior researchers** in **adopting LLM/ML technologies**, aligning with **industry best practices**.

CS Coach at Global Learning Centre:

- Mentored over 100 undergrad CS students and provided one-on-one lessons to students from diverse backgrounds.
- Assisted with assignments, projects, and course materials; guided students on career roadmaps. (Aug 2023 – Apr 2025)

Teaching Assistant & Lab Instructor: Worked on **11 different** undergrad and graduate courses.

(Jan 2023 – Apr 2025)

D.P.O. International Sdn Bhd - (Multinational e-commerce business company)

(Sep 2021 – Jan 2023)

- **Developed and maintained scalable ASP .NET applications** using **C#** and **Microsoft SQL Server**.
- **Optimized transaction processing** by integrating microservices, resulting in a **37% reduction in processing time**.
- **Designed and built RESTful APIs**, ensuring seamless integration between **microservices** for **real-time transactions**.
- **Implemented RabbitMQ** to improve **communication** between **distributed systems**, enhancing data consistency.
- Partnered with **vendors** to add **Elasticsearch** and **RabbitMQ**, improving **search functionality** and **system reliability**.
- Conducted **software testing and research** on emerging technologies for continuous improvement and innovation.

DHL Asia Pacific Shared Service Centre

(Feb 2021 – Aug 2021)

Software Engineer (Intern) at BPO Department

Kuala Lumpur, Malaysia

- **Developed internal software applications** using **ASP .Net Web Form** and **MVC framework** with **C#**.
- **Converted legacy Excel VBA macros** to scalable **.Net web applications**, enhancing functionality and maintainability.
- **Resolved IT tickets** and **debugged existing applications**, ensuring minimal downtime and system reliability.

Academic and Personal Projects:

Telecom Question Answering with Retrieval-Augmented Generation: Developed a novel **telecom-specific multi-modal RAG** pipeline combining **finetuned embedding models and LLMs** to enhance **QA** accuracy on **3GPP documents**. Achieved a **47% improvement in accuracy** compared to the baseline. Used **Colbert, Matryoshka Representative Learning, Phi2, ChromaDB** for the **RAG** pipeline. **Fine-tuned** a number of **open-source LLMs** on telecom data.

LLM-Driven Telecom Incident Management Classification: Built a multi-label classification pipeline using **RAG** to predict four incident ticket types for IMTs from a real Canadian telecom dataset. Implemented a **BERT-based** model with **Bayesian feature-selection**, boosting minority-class **F1 by over 14%**. Leveraged a **Phi-2** model to achieve a further **12.7% F1 gain**.

Hierarchical Network Intrusion Detection System: Built a hierarchical **IDS** leveraging a **CNN-based classifier** to distinguish “large” vs. “small” attack groups, then applying a **ResNet-style CNN** and **Random Forest** to each group, that achieved **≥ 92%** overall **F1 score**, improving minority attack group’s **F1 score by 22.5%** on a real SD-WAN dataset.

Education:

University of Regina

(Jan 2023- May 2025)

Degree: Master of Computer Science – Thesis Route

Regina, Saskatchewan

GPA: 88.40%

International Islamic University Malaysia

(Jun 2017- Aug 2021)

Degree: Bachelor of Computer Science

Kuala Lumpur, Malaysia

CGPA: 3.62 / 4.00

Awards & Scholarships:

- **Fully Funded Graduate Research Scholarship & Departmental Grant** (Winter 2023 – Winter 2025)
- **Saskatchewan Innovation and Excellence Graduate Scholarship** - Fall 2023
- **GSA Travel Award & FGSR Graduate Student Travel Award** to attend **IEEE GLOBECOM 2024** in **Cape Town**.

Extra-Curricular Activities:

- **Vice President - Culture:** University of Regina Bangladeshi Students’ Association (Sep 2023 – Oct 2024)
- **International Peer Advisor** at UR International. Organized 3 orientation events, volunteered at the airport welcome booth for 4 semesters, and volunteered at 3 other international events. (Jan 2024 – Apr 2025)
- **Volunteer at IEEE Canada:** IEEE Conference CCECE 2023

References: Available upon request.